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電子物理學系
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諧波產生效應的影響

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摘 要

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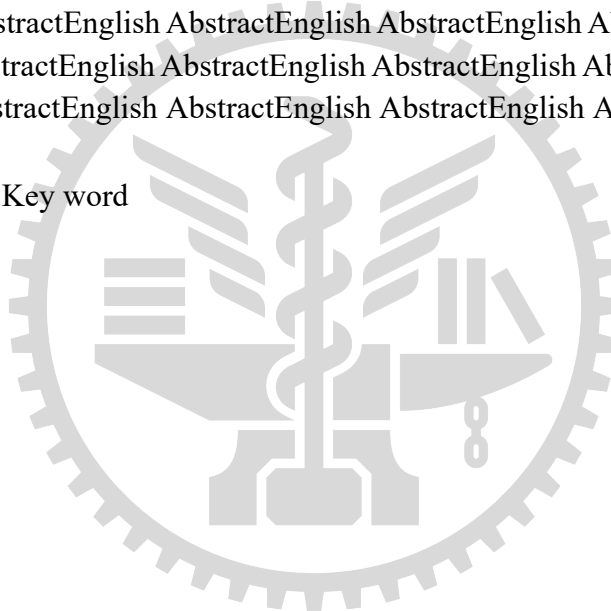
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Abstract

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Contents

Chinese Abstract	i
English Abstract	ii
Contents	iii
Figure Captions	iv
Table Captions	v
Chapter1 Introduction	1
1.1 Introduction	1
Chapter2 Twisted-Light with Orbital Angular Momentum.	2
2.1 Paraxial Helmholtz equation	2
2.1.1 Wave Equations	2
2.1.2 Gauge Freedom	3
2.1.3 Paraxial Helmholtz Equation	4
2.2 Twisted Light in Laguerre-Gaussian mode	5
2.2.1 Real space representation under the Lorenz gauge	5
2.2.2 Angular spectrum representation under the Lorenz gauge	5
2.2.3 Real space representation under the Coulomb gauge	6
2.2.4 Angular spectrum representation under the Coulomb gauge	7
2.2.5 Vector Vortex Beams	7
Chapter3 Light-Matter Interaction.	8
3.1 Optical transition with Twisted Light	8
3.1.1 Optical transition with Twisted Light	8
Chapter4 Summary and outlook	9
References	10

Figure Captions



Table Captions



Chapter 1 Introduction

1.1 Introduction



Chapter 2 Twisted-Light with Orbital Angular Momentum

2.1 Paraxial Helmholtz equation

2.1.1 Wave Equations

The Maxwell's equations describe how **electric field** $E(\mathbf{r}, t)$ and **magnetic field** $B(\mathbf{r}, t)$ behave in given position $\mathbf{r}$ and time t , including Laguerre-Gaussian beam. In **free space**, the charge density ρ and current density $\mathbf{J}$ are absent, making the Maxwell's equations four homogeneous differential equations with vacuum permittivity ϵ_0 and vacuum permeability μ_0 and speed of light in vacuum $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$. In electrodynamics scheme, the Maxwell's equations are

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t) = \frac{\rho}{\epsilon_0} = 0, \quad (2.1)$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0, \quad (2.2)$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial}{\partial t} \mathbf{B}(\mathbf{r}, t), \quad (2.3)$$

$$\nabla \times \mathbf{B}(\mathbf{r}, t) = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}(\mathbf{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}(\mathbf{r}, t). \quad (2.4)$$

Where $\nabla \equiv \frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}} + \frac{\partial}{\partial z} \hat{\mathbf{z}}$ is the partial derivatives along each of the three coordinate directions. In addition, the four equations can be generalized in potential form and will be used in the light-matter interaction theory latter. First, the divergenceless of Eq. (2.2) introduce a **vector potential** $\mathbf{A}(\mathbf{r}, t)$:

$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (2.5)$$

Second, the presence of $\mathbf{A}(\mathbf{r}, t)$ and the Faraday's law (Eq. (2.3)) gives a quantity that vanishes with its curl and gives rise to a **scalar potential** $\phi(\mathbf{r}, t)$:

$$\nabla \phi(\mathbf{r}, t) = \left(\mathbf{E}(\mathbf{r}, t) + \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}, t) \right). \quad (2.6)$$

Then, in terms of $\mathbf{A}(\mathbf{r}, t)$ and $\phi(\mathbf{r}, t)$, then the electric field can be written as

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \phi(\mathbf{r}, t) - \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}, t) \quad (2.7)$$

which will certainly satisfies both Eq. (2.2) and Faraday's law (Eq. (2.3)). Finally, the Gauss's law (Eq. (2.1)) and the Ampère's law with Maxwell's correction (Eq. (2.4)) can be rewritten as

two wave equations for the electromagnetic potentials

$$\nabla^2 \phi(\mathbf{r}, t) + \frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}(\mathbf{r}, t)) = 0, \text{ and} \quad (2.8)$$

$$\nabla^2 \mathbf{A}(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}(\mathbf{r}, t) - \nabla \left(\nabla \cdot \mathbf{A}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial}{\partial t} \phi(\mathbf{r}, t) \right) = 0. \quad (2.9)$$

The two equation remain all the information in Maxwell's equations.

2.1.2 Gauge Freedom

Further simplification of the wave equations Eq. (2.8) and Eq. (2.9) can be achieved by exploiting the gauge freedom of the vector and scalar potentials, $\mathbf{A}(\mathbf{r}, t)$ and $\phi(\mathbf{r}, t)$. Below, we introduce two commonly used gauges: the **Coulomb gauge** and the **Lorenz gauge**. The former is widely adopted in light-matter interaction theory, while the latter is employed here for the derivation of Laguerre-Gaussian beams. Superscripts C and L are used in the following derivations to distinguish quantities defined in the Coulomb and Lorenz gauges, respectively. The gauges conditions are given as

$$\text{Coulomb gauge condition : } \nabla \cdot \mathbf{A}^C(\mathbf{r}, t) = 0, \quad (2.10)$$

$$\text{Lorenz gauge conditon : } \nabla \cdot \mathbf{A}^L(\mathbf{r}, t) = -\frac{1}{c^2} \frac{\partial}{\partial t} \phi^L(\mathbf{r}, t). \quad (2.11)$$

Both gauges can simplify the wave equations Eq. (2.8) and Eq. (2.9). In Coulomb gauge, the wave equations turn into

$$\nabla^2 \phi^C(\mathbf{r}, t) = 0, \text{ and} \quad (2.12)$$

$$\nabla^2 \mathbf{A}^C(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^C(\mathbf{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \phi^C(\mathbf{r}, t). \quad (2.13)$$

Where Eq. (2.12) has trivial solution $\phi^C(\mathbf{r}, t) = 0$ and Eq. (2.13) can be rewritten as $\nabla^2 \mathbf{A}^C(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^C(\mathbf{r}, t) = 0$. In Lorenz gauge, the wave equations Eq. (2.8) and Eq. (2.9) turn into

$$\nabla^2 \phi^L(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \phi^L(\mathbf{r}, t) = 0, \text{ and} \quad (2.14)$$

$$\nabla^2 \mathbf{A}^L(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^L(\mathbf{r}, t) = 0. \quad (2.15)$$

The physical quantity $\mathbf{E}(\mathbf{r}, t)$ can be related to vector potentials in different gauges through the relation

$$\mathbf{E}(\mathbf{r}) = i\omega \mathbf{A}^C(\mathbf{r}) = i\omega \left(\mathbf{A}^L(\mathbf{r}) + \frac{\nabla (\nabla \cdot \mathbf{A}^L(\mathbf{r}))}{q_0^2} \right) \quad (2.16)$$

The derivation of this relation has adopt the ansatz of **monochromatic wave** (for electric field $\mathbf{E}(\mathbf{r}, t) = \text{Re} [\mathbf{E}(\mathbf{r}) e^{-i\omega t}]$, scalar potentials $\phi^L(\mathbf{r}, t) = \text{Re} [\phi^L(\mathbf{r}) e^{-i\omega t}]$, vector potential $\mathbf{A}^C(\mathbf{r}, t) = \text{Re} [\mathbf{A}^C(\mathbf{r}) e^{-i\omega t}]$, and $\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r}) e^{-i\omega t}]$ with ω angular frequency) and derived from Eq. (2.7) under corresponding gauge condition. Where $q_0 = \omega/c$ is wavenumber of light

2.1.3 Paraxial Helmholtz Equation

In principle, to obtain the solution in vector potential form we can keep on the derivation with either Eq. (2.13) or Eq. (2.15), in conclusion we will use **Lorenz gauge** with Eq. (2.15) and leave the reason at the end of this section. As the first step, the **monochromatic wave** ansatz ($\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r})e^{-i\omega t}]$) turns Eq. (2.15) into vector Helmholtz equation

$$\nabla^2 \mathbf{A}^L(\mathbf{r}) + q^2 \mathbf{A}^L(\mathbf{r}) = 0. \quad (2.17)$$

Next, the **transverse wave** ansatz ($\mathbf{A}^L(\mathbf{r}) = \hat{\mathbf{e}} A^L(\mathbf{r})$ with polarization vector of unit modulus $\hat{\mathbf{e}}$) can be rewritten Eq. (2.17) as scalar Helmholtz equation

$$\nabla^2 A^L(\mathbf{r}) + q^2 A^L(\mathbf{r}) = 0. \quad (2.18)$$

We further assume that the rays have only small inclinations with respect to the optical axis, a condition referred to as the **paraxial approximation** in geometric optics. In wave optics, this corresponds to the angular spectrum of a paraxial light beam having dominant plane-wave components lie close to the beam propagation direction, which is chosen to be along the z axis. As a result, the largest component of a wavevector $\mathbf{q}_0$ lies along the $\hat{\mathbf{z}}$ direction with

$$q_z = \sqrt{q_0^2 - q_{\parallel}^2} = q_0 \sqrt{1 - \frac{q_{\parallel}^2}{q_0^2}} \approx q_0 - \frac{q_{\parallel}^2}{2q_0}, \quad (2.19)$$

where the value of in-plane wave vector $q_{\parallel} = \sqrt{q_x^2 + q_y^2}$. With the approximation, we consider the **paraxial wave** ansatz ($A^L(\mathbf{r}) = U(\mathbf{r})e^{iq_0 z}$) and substitute into scalar Helmholtz equation (Eq. (2.18)) gives

$$\nabla_{\parallel}^2 U(\mathbf{r}) + \frac{\partial^2}{\partial z^2} U(\mathbf{r}) + 2iq_0 \frac{\partial}{\partial z} U(\mathbf{r}) = 0 \quad (2.20)$$

Where $\nabla_{\parallel}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is the transverse Laplace operator. The ansatz features a slowly varying spatial distribution ($U(\mathbf{r})$) along z direction comparing to primary variation ($e^{iq_0 z}$), thus holds the following relations $\left| \frac{\partial^2}{\partial z^2} U(\mathbf{r}) \right| \ll \left| \nabla_{\parallel}^2 U(\mathbf{r}) \right|$, and $\left| \frac{\partial^2}{\partial z^2} U(\mathbf{r}) \right| \ll q_0 \left| \frac{\partial}{\partial z} U(\mathbf{r}) \right|$. Neglecting the second-order derivative with respect to z constitutes the paraxial approximation and results in the paraxial Helmholtz equation

$$\nabla_{\parallel}^2 U(\mathbf{r}) + 2iq_0 \frac{\partial}{\partial z} U(\mathbf{r}) = 0. \quad (2.21)$$

Note that paraxial wave ansatz will not hold the relations in Coulomb gauge since the gauge condition (Eq. (2.10)) gives

$$\nabla \cdot \mathbf{A}^C(\mathbf{r}, t) = \hat{\mathbf{e}} \cdot \nabla A^C(\mathbf{r}, t) = 0. \quad (2.22)$$

For a transverse wave whose polarization vector only have x and y components, Eq. (2.22) simplifies to $\hat{\mathbf{e}} \cdot \nabla_{\parallel} A^C(\mathbf{r}, t) = 0$, which forces zero transverse variation along the in-plane direction ($\nabla_{\parallel} = \frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}}$). Thus violate our ansatz that $U(\mathbf{r})$ is slowly varying along z direction, unless a non-zero z component is allowed to exist to satisfy the divergence free condition.

2.2 Twisted Light in Laguerre-Gaussian mode

2.2.1 Real space representation under the Lorenz gauge

The paraxial Helmholtz equation (Eq. (2.21)) can be solved under various coordinate systems and yields different solutions such as **Hermite-Gaussian** beam, **Laguerre-Gaussian** (LG) beam, and **Bessel** beams, etc. In the twisted Light theory we take the widely used mode, LG beams, solved from the paraxial Helmholtz equation in cylindrical coordinates

$$\left(\frac{1}{\rho} \frac{\partial}{\partial \rho} + \frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial^2}{\partial \phi^2} + 2iq \frac{\partial}{\partial z} \right) U(\rho, \phi, z) = 0 \quad (2.23)$$

Where the position vector $\mathbf{r} = (x, y, z) = (\mathbf{r}_{\parallel}, z) = (\rho, \phi, z)$, and $\mathbf{r}_{\parallel}$ is the in-plane position vector which contain ρ radial coordinate and ϕ azimuthal angle. By considering **long Rayleigh-length conditionn**, the LG beam is than solved as

$$\mathbf{A}_{\mathbf{q}_0}^{\hat{\epsilon}, \ell, p; L}(\mathbf{r}) = \hat{\epsilon} A_0^{\text{LG}} u_{\ell p}(\mathbf{r}_{\parallel}) e^{iq_0 z}, \quad (2.24)$$

where the subscripted integer $\ell \in \mathbb{Z}$ is the index of azimuthal mode and refer as **topological charge**, subscripted integer $p \geq 0$ is the index of radial mode, $\mathbf{q}_0 = q_0 \hat{z}$ is the wavevector specified to be along the direction of light propagation, and A_0^{LG} is the constant amplitude of the vector potential. The spatial distribution is expressed as

$$U(\rho, \phi, z) \approx u_{\ell p}(\mathbf{r}_{\parallel}) = f_{\ell p}(\rho) e^{i\ell \phi}, \quad (2.25)$$

and follows

$$f_{\ell p}(\rho) = \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} \exp \left(-\frac{\rho^2}{W_0^2} \right), \quad (2.26)$$

where $L_p^{|\ell|}(x) = \sum_{m=0}^p (-1)^m [(|\ell| + p)! / (p - m)! (|\ell| + m)! m!] x^m$ is the **generalized Laguerre polynomial** with the variable $x = \left(\frac{2\rho^2}{W_0^2} \right)$, and parametrized by the beam waist W_0 , $\rho = \sqrt{x^2 + y^2}$, $\phi = \tan^{-1}(y/x)$.

2.2.2 Angular spectrum representation under the Lorenz gauge

Since the light-matter interaction theory include LG beams in angular spectrum representation, we need to describe the vector potential in terms of plane-wave basis

$$\mathbf{A}(\mathbf{r}) = \sum_{\mathbf{q}} \mathcal{A}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}}, \quad (2.27)$$

where $\mathcal{A}^L(\mathbf{q})$ is the (complex) amplitude with each $e^{i\mathbf{q}\cdot\mathbf{r}}$ as plane-wave basis. The LG beam in Eq. (2.24) can obtain this form through **two-dimensional Fourier transformation**

$$u_{\ell p}(\mathbf{r}_{\parallel}) = \frac{1}{(2\pi)^2} \int d^2\mathbf{q}_{\parallel} F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel}\cdot\mathbf{r}_{\parallel}}, \quad (2.28)$$

and determine $\mathcal{A}^L(\mathbf{q}_{\parallel})$ with the **inverse transformation**

$$F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) = \int d^2\mathbf{r}_{\parallel} u_{\ell p}(\mathbf{r}_{\parallel}) e^{-i\mathbf{q}_{\parallel}\cdot\mathbf{r}_{\parallel}}, \quad (2.29)$$

where the in-plane wave vector $\mathbf{q}_{\parallel} = (q_{\parallel}, \phi_{q_{\parallel}})$. Since the spatial distribution has the form $u_{\ell p}(\mathbf{r}_{\parallel}) = f_{\ell p}(\rho) e^{i\ell\phi}$, the amplitude for plane-wave basis can be solved analytically as $F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) = (-1)^{|\ell|} 2\pi \mathcal{H}_{|\ell|}[f_{\ell p}(\rho)] e^{i\ell\phi_{q_{\parallel}}}$, where the **Hankel transformation** $\mathcal{H}_{|\ell|}[f_{\ell p}(\rho)] = \int_0^\infty d\rho \rho f_{\ell p}(\rho) J_{|\ell|}(q_{\parallel}\rho)$ of order $|\ell|$ and $J_{|\ell|}(q_{\parallel}\rho)$, the **Bessel function of the first kind** of order $|\ell|$. Further approximately regard $\mathbf{q}_{\parallel}$ as the vector for 2D material in reciprocal space so that Eq. (2.28) can be discretized and the LG beams in angular spectrum representation become

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}) = \hat{\varepsilon} \sum_{\mathbf{q}_{\parallel}} \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel}\cdot\mathbf{r}_{\parallel}} e^{iq_0 z}, \quad (2.30)$$

where the amplitude function of LG beam is given by

$$\mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) = \tilde{F}_{\ell p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}}, \quad (2.31)$$

and the radial function $\tilde{F}_{\ell p}(q_{\parallel}) = \frac{A_0^{\text{LG}}}{\Omega} (-1)^{|\ell|} 2\pi \mathcal{H}_{|\ell|}[f_{\ell p}(\rho)]$. This study focuses on the fundamental radial mode $p = 0$ for LG beams, which have explicit form in real space as

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; L}(\mathbf{r}) = \hat{\varepsilon} \frac{A_0^{\text{LG}}}{\sqrt{|\ell|!} \pi/2} \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} \exp\left(-\frac{\rho^2}{W_0^2}\right) \exp[i(\ell\phi + q_0 z)], \quad (2.32)$$

whereas in angular spectrum representation

$$\mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) = \left(\frac{\pi A_0^{\text{LG}} W_0^2 (-1)^{|\ell|}}{\Omega \sqrt{|\ell|!} \pi/2} \right) \left[\left(\frac{q_{\parallel} W_0}{\sqrt{2}} \right)^{|\ell|} \exp\left(-\frac{q_{\parallel}^2 W_0^2}{4}\right) \right] \exp(i\ell\phi_{q_{\parallel}}) \quad (2.33)$$

add some figure

2.2.3 Real space representation under the Coulomb gauge

While the light-matter interaction theory include LG beams in Coulomb gauge, The LG beams are derived in Lorenz gauge. Therefore, it is vital to convert our solutions into Coulomb gauge. In previous section we already have Eq. (2.16) that implicit the connection between

vector potentials in different gauges, which gives us

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; C}(\mathbf{r}) = \mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}) + \frac{\nabla (\nabla \cdot \mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}))}{q_0^2}. \quad (2.34)$$

For LG beams in fundamental radial mode $p = 0$

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r}) \approx A_0^{\text{LG}} \left[f_{\ell p=0}(\rho) e^{i\ell\phi} \hat{\varepsilon} + \frac{i}{\sqrt{2}q_0\rho} \left((|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right) f_{\ell p=0}(\rho) e^{i(\sigma+\ell)\phi} \hat{z} \right] e^{iq_0 z} \quad (2.35)$$

$$\hat{\varepsilon} = \frac{1}{\sqrt{2}}(\hat{x} + i\sigma\hat{y})$$

2.2.4 Angular spectrum representation under the Coulomb gauge

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r}) = \frac{1}{(2\pi)^2} \int d^2\mathbf{q}_{\parallel} \mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} \quad (2.36)$$

$$\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) = \left[\hat{\varepsilon} - \left(\frac{|\hat{\mathbf{q}}||\hat{\mathbf{q}}_{\parallel}|}{q_0^2} \right) (\hat{\varepsilon} \cdot \hat{\mathbf{q}}_{\parallel}) \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.37)$$

$$= \left[\hat{\varepsilon} - \left(\frac{|\mathbf{q}|}{q_0} \right)^2 \frac{\sin\theta_{q_{\parallel}}}{\sqrt{2}} e^{i\sigma\phi_{q_{\parallel}}} \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.38)$$

$$\approx \left[\hat{\varepsilon} - \frac{\sin\theta_{q_{\parallel}}}{\sqrt{2}} e^{i\sigma\phi_{q_{\parallel}}} \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.39)$$

$$\quad (2.40)$$

$$\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) \approx \tilde{F}_{\ell p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}} \hat{\varepsilon} - \frac{\sin\theta_{q_{\parallel}}}{\sqrt{2}} \tilde{F}_{\ell p}(q_{\parallel}) e^{i(\sigma+\ell)\phi_{q_{\parallel}}} \hat{z} \quad (2.41)$$

2.2.5 Vector Vortex Beams

Chapter 3 Light-Matter Interaction

3.1 Optical transition with Twisted Light

3.1.1 Optical transition with Twisted Light

The transition rate of an exciton state excited by time-dependent perturbation theory, the time-dependent exciton state under a weak photoexcitation can be expressed as



Chapter 4 Summary and outlook



國立陽明交通大學學位論文發表形式確認書

本人 姚述涵 依據「國立陽明交通大學著作彙編之學位論文應行注意事項」第四點第一項規定，於 114 年 7 月 10 日與指導教授討論並取得論文發表形式之共識，本人學位論文為：

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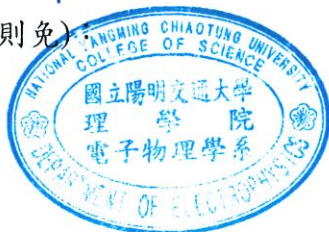
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